

Crime Analytics & Crime Hotspot Prediction System

A Machine Learning-Based Crime Intelligence & Risk Mapping Platform

This is a strong **B.Sc. Data Science ML** project because it combines:

- Data preprocessing
- Exploratory Data Analysis
- Geospatial analytics
- Machine learning
- Clustering
- Classification
- Time-series analysis
- Interactive visualization
- Explainable AI

The goal is **not to predict individual crimes or identify criminals**, but to analyze historical crime patterns and estimate **crime risk levels for locations and time periods** to support better resource planning.

1. Project Overview

Problem Statement

Crime data is usually available as large historical records containing:

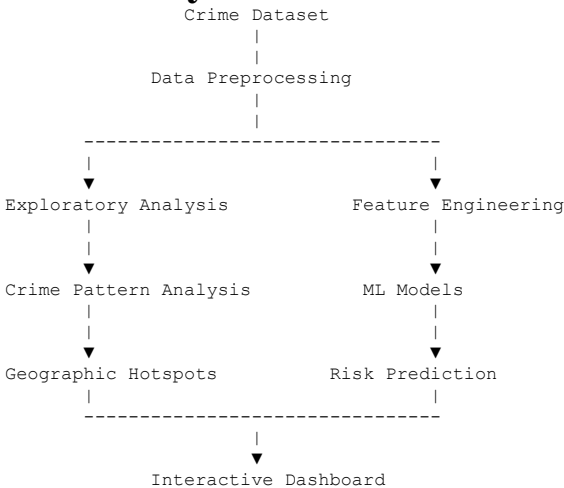
- Crime type
- Location
- Time
- Victim information
- Case status

Manually analyzing these records is difficult.

This project builds a system that:

1. Identifies crime hotspots.
2. Finds crime patterns.
3. Predicts crime risk levels.
4. Analyzes factors influencing crime occurrence.
5. Provides interactive maps and dashboards.

2. Final System Architecture



3. Dataset Collection

Option 1: Kaggle Datasets

Examples:

- Chicago Crime Dataset
- Los Angeles Crime Dataset
- San Francisco Crime Dataset

These are large and detailed.

Option 2: Indian Crime Data

Sources:

- NCRB (National Crime Records Bureau)
- State police open datasets
- Government open data portals

4. Dataset Structure

Example:

Column	Description
ID	Crime record ID
Date	Date of crime
Time	Time of crime
City	Location
Latitude	GPS latitude
Longitude	GPS longitude
Crime Type	Theft, Assault, Robbery
Weapon	Weapon used
Victim Age	Age
Victim Gender	Gender
Location Type	Street, Home, Park
Arrest	Arrest made
Case Status	Closed/Open

5. Project Modules

MODULE 1: Data Collection & Cleaning

Tasks

Missing Value Handling

Examples:

Missing location:

Latitude = NULL

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Remove or estimate using nearby records

Date Processing

Convert:

01/02/2024 18:30

into:

Day = Monday

Month = February

Year = 2024

Hour = 18

Season = Winter

Duplicate Removal

Remove repeated crime reports.

Standardization

Example:

Before:

Theft

THEFT

theft

After:

THEFT

MODULE 2: Exploratory Data Analysis

This is a major part.

A. Crime Distribution Analysis

Questions:

- Which crimes happen most?
- Which areas have more crime?
- Which year had maximum crime?

Visualizations:

Crime Type Distribution

Chart:

Bar Chart

Theft
Assault
Robbery



Crime Trend Over Years

Line chart:

2019	----
2020	-----
2021	-----
2022	-----

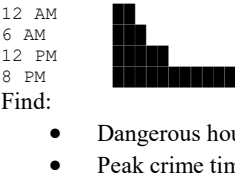
B. Time Analysis

Analyze:

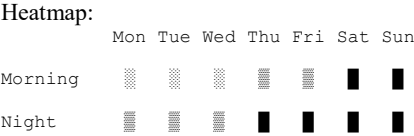
Hourly Pattern

Example:

Crime Frequency



Day Analysis



C. Location Analysis

Analyze:

- Most dangerous areas
- Crime density
- Crime type by location

MODULE 3: Geospatial Analysis

This is where the project becomes unique.

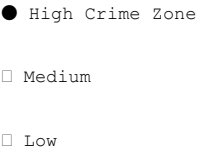
Crime Heatmap

Using:

- Folium
- Plotly Maps

Example:

City Map



Hotspot Detection

Algorithms:

DBSCAN ★ Recommended

Why?

Crime locations are geographic points.

Input:

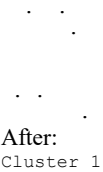
Latitude
Longitude

DBSCAN identifies clusters.

Example:

Before:

Crime Points:





Cluster 2



Parameters:

eps = distance radius

min_samples = minimum crimes

Output:

Hotspot ID

Location

Crime Density

MODULE 4: Feature Engineering

Create ML features.

Time Features

Hour

Day

Month

Weekend

Holiday

Season

Location Features

Latitude

Longitude

Crime Density

Previous crimes nearby

Crime History Features

Example:

Last 30 days:

Crime count = 45

Last 90 days:

Crime count = 130

Weather Features (Optional)

Use weather API.

Add:

Temperature

Rain

Humidity

Analyze:

Does weather affect crime?

MODULE 5: Machine Learning Models

You can build two ML components.

MODEL 1: Crime Risk Classification ★ ★ ★ ★ ★

Objective

Predict:

Location + Time

↓

Crime Risk Level

Output:

Low Risk

Medium Risk

High Risk

Features

Input:
Latitude

Longitude

Hour

Day

Month

Previous crime count

Crime type

Weather

Algorithms

Baseline

Logistic Regression

Main Models

1. Random Forest ★
2. XGBoost
3. LightGBM

Evaluation

Metrics:
Accuracy
Precision
Recall
F1 Score
Confusion Matrix

MODEL 2: Crime Count Forecasting

Predict:
"How many crimes may happen next week?"

Models

Traditional

- ARIMA

ML

- Random Forest Regression
- XGBoost Regression

Deep Learning (Optional)

- LSTM

Features:
Previous crime count

Month

Season

Holiday

Location

Metrics:
MAE
RMSE
MAPE

MODULE 6: Explainable AI

Use:

SHAP

Answer:

"Why is this area predicted high risk?"

Example output:

Risk Prediction: HIGH

Reasons:

+ Previous crime density 45%
+ Night time 25%
+ Weekend 15%
+ Location type 10%

Visualization:

MODULE 7: Dashboard Development

Use:
Streamlit

Page 1: Crime Overview

Cards:
Total Crimes

Crime Rate

Most Common Crime

Highest Risk Area

Current Risk Level

Page 2: Crime Map

Interactive map:
Features:
 • Heatmap
 • Cluster markers
 • Filters

Filters:
Crime Type

Year

Area

Risk Level

Page 3: Crime Analysis

Charts:
 • Crime category distribution
 • Monthly trends
 • Hourly trends
 • Day-wise analysis

Page 4: Hotspot Detection

Show:
Cluster 1

Area:
Downtown

Crime Density:
High

Main Crime:
Theft

Page 5: Prediction

User inputs:
Location

Date

Time

Crime Type
Output:
Risk:

HIGH

Probability:

82%

Important Factors:

Night time
Previous crimes
Location

Page 6: City Comparison

Compare:
City A

vs

City B

Metrics:

- Crime rate
- Crime types
- Trend
- Risk score

8. Advanced Features (Optional)

1. Crime Pattern Mining

Use:
Apriori Algorithm
Find:
Night + Weekend + Area X
→ Theft probability high

2. Social Media Sentiment

Analyze local news/social posts.
NLP:

- Sentiment analysis
- Topic modeling

3. Automated Crime Report Generator

Generate:
Monthly Crime Intelligence Report

Top hotspots:
1.
2.
3.

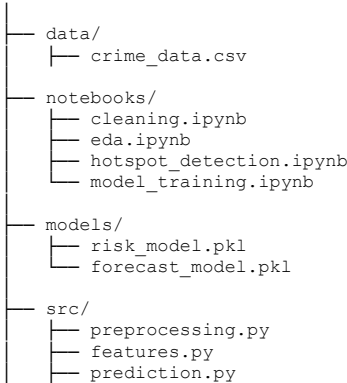
Recommendations:
...

9. Technology Stack

Task	Technology
Programming	Python
Data Processing	Pandas
Visualization	Plotly, Seaborn
Maps	Folium
ML	Scikit-learn
Boosting	XGBoost
Clustering	DBSCAN
Explainability	SHAP
Dashboard	Streamlit
Database	SQLite/PostgreSQL

10. Project Folder Structure

CrimeAnalyticsSystem/



```
├── visualization.py
├── dashboard/
│   └── app.py
├── reports/
│   └── final_report.pdf
└── requirements.txt
```

11. Timeline (8 Weeks)

Week	Work
1	Collect dataset + cleaning
2	EDA + visualization
3	Feature engineering
4	Hotspot detection using DBSCAN
5	Build ML models
6	Model evaluation + SHAP
7	Streamlit dashboard
8	Testing + documentation

Final Project Output

A complete system:

- ✓ Crime trend analysis
- ✓ Crime hotspot detection
- ✓ Interactive crime maps
- ✓ ML-based risk prediction
- ✓ Crime forecasting
- ✓ Explainable AI
- ✓ Dashboard deployment